
Advanced Topics in Machine Learning - PA1

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Abstract

This assignment explores visual biases, domain adaptation (DA), domain generalization (DG), and open-set recognition (OSR). ResNet, Vision Transformer, and CLIP are compared under controlled image interventions to examine cue dependence and representation stability. DAN, DANN, CDAN, and SAM are evaluated for domain transfer, while novelty scores, augmentation, and PROSER are examined for unknown rejection. The analysis considers how representations preserve class information, respond to distribution shifts, and support rejection of unfamiliar inputs.

1 Introduction

Image classifiers typically assume that training and test data share similar distributions and the same set of classes. In practice, objects may appear with different colors, textures, or spatial arrangements; images may come from an unseen domain; and inputs may belong to unfamiliar classes. These changes can expose weaknesses that clean classification accuracy does not capture.

This assignment examines these challenges through four tasks. Controlled interventions on STL-10 investigate visual biases in ResNet, ViT, and CLIP. Experiments on PACS compare adaptation with access to unlabeled target images against generalization using source domains alone. CIFAR-10 and CIFAR-100 experiments assess whether classifiers can distinguish known objects from unknown classes.

The experiments address the following research questions:

- How do shape, texture, color, and spatial structure influence predictions, and do prediction changes agree with changes in feature representations?
- Does access to unlabeled target data improve recognition through domain alignment, and when does alignment damage class-discriminative information?
- Can source-domain alignment or local parameter stability improve recognition on an unseen domain?
- How does semantic similarity affect unknown rejection, and do novelty scores, augmentation, or placeholder learning improve rejection while preserving known-class accuracy?

Across these tasks, the central question is what a robust representation should preserve, which variations it should ignore, and when an input should be rejected rather than assigned to a known class.

2 Task 1: Visual Cues and Representation Stability

Setup: ResNet-50, ViT-B/16, and CLIP were frozen and evaluated on the same balanced subset of 500 STL-10 test images, with 50 images per class. CLIP used both a trained linear head and zero-shot classification. The additional color intervention was hue rotation of $+0.25$.

All models received identical interventions on shared 224×224 images before model-specific normalization, using seed 6304. Translation used reflection padding at 0, 8, 16, and 32 pixels. Patch shuffling used a fixed non-identity permutation of a 4×4 grid per image, with each patch measuring 56×56 pixels.

For shape–texture evaluation, VGG-19 style transfer generated conflicts across cat/dog, deer/horse, bird/monkey, car/truck, and airplane/ship. Acceptance required a recognizable content outline and visible transferred surface pattern on the object. Visual review retained 200 conflicts, with 20 per direction, and rejected 375 candidates lacking visible object-texture transfer. 200 Adam iterations were used with a learning rate of 0.02, content weight of 1, and a style weight of 10^6 .

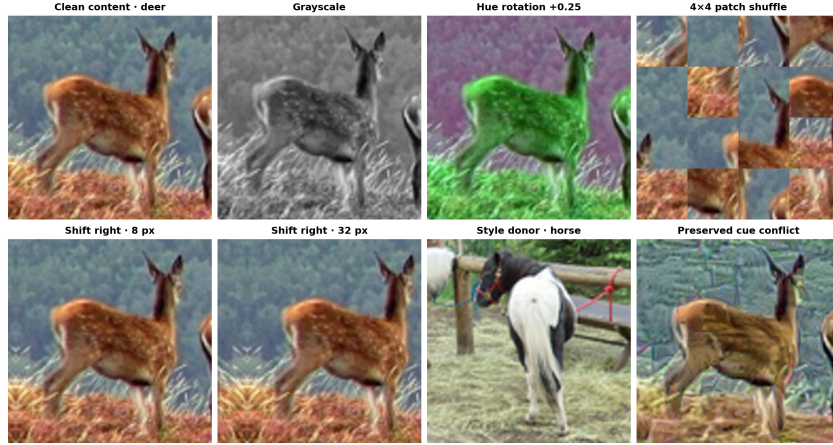


Figure 1: The same deer image under grayscale, hue rotation, patch shuffling, and translation, alongside the horse style reference and accepted deer–horse cue conflict.

Color and shape–texture reliance (RQ1): Clean accuracy was similar across models since the difference between the highest and lowest accuracy was only 0.8 percentage points.

Grayscale reduced accuracy by 1.6–2.4 percentage points, while hue rotation caused larger losses, particularly for ResNet and zero-shot CLIP. This can imply that color contributed to decisions. Grayscale preserves both shape and substantial texture, so grayscale robustness alone does not mean that the model relies on shape.

Table 1: Clean metrics and intervention performance (%). Transformed accuracy is followed by prediction consistency in parentheses. C and Z denote CLIP head and zero-shot CLIP.

Metric / condition	ResNet	ViT	C	Z
Clean accuracy	97.6	97.4	97.2	96.8
Clean macro-F1	97.6	97.4	97.2	96.8
Clean confidence	89.8	95.9	14.2	94.3
Grayscale	95.8	95.4	95.6	94.4
	(96.6)	(95.0)	(94.8)	(95.4)
Hue rotation	92.8	95.0	94.6	92.6
	(94.0)	(94.8)	(94.0)	(93.2)
Patch shuffle	87.4	90.6	79.2	77.4
	(89.8)	(91.8)	(79.0)	(77.2)

ViT showed the strongest aggregate shape preference, followed by CLIP. ResNet made comparatively more texture decisions, but still selected shape more often than texture.

Cue preference also depended on direction since the CLIP head selected shape on all airplane-to-ship conflicts, but its reverse-direction shape bias was 46.7%, with 75% coverage.

Table 2: Cue-conflict decisions on 200 images. S/T/O denotes shape/texture/other counts. Shape bias excludes other decisions; coverage includes shape and texture decisions.

Model	S/T/O	Shape bias	Coverage
ResNet	108/77/15	58.4%	92.5%
ViT	154/37/9	80.6%	95.5%
CLIP head	137/47/16	74.5%	92.0%
CLIP zero-shot	131/48/21	73.2%	89.5%

Spatial Structure (RQ2): At 32-pixel translation, accuracy losses were only 0.25–1.20 percentage points and prediction consistency exceeded 97% for every model. ResNet had the highest consistency, at 98.65%.

Patch shuffling caused larger accuracy losses for all the models and increased losses by 10.2, 6.8, 18.0, and 19.4 percentage points for ResNet, ViT, CLIP head, and zero-shot CLIP, respectively. CLIP was the most sensitive to disrupted spatial organization.

ViT’s strong shape preference and shuffle robustness are compatible since cue conflicts test preference between content and transferred style, whereas shuffling retains substantial evidence within each patch.

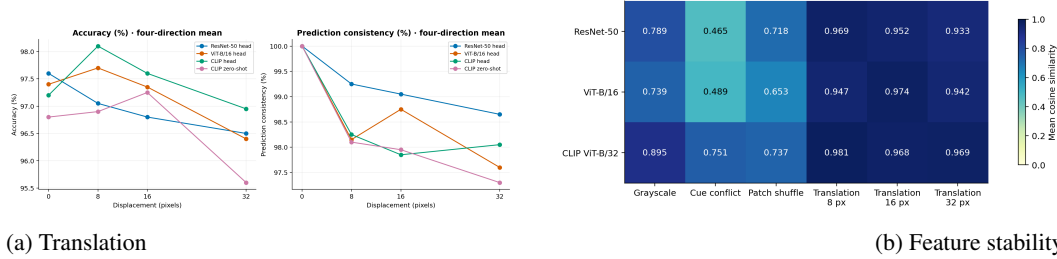


Figure 2: Prediction and representation stability. Translation results average four directions. The heatmap reports mean cosine similarity.

Representations versus Predictions (RQ3): Cosine similarity paired each transformed representation with its clean counterpart, using the content image for cue conflicts. Similarity was measured in the original feature space.

Translation preserved both features and predictions, but this relationship was not universal. ViT had the lowest patch-shuffle cosine similarity, 0.653, yet the smallest accuracy loss. A shuffled truck retained its correct ViT prediction despite cosine similarity of 0.274. Conversely, a translated deer changed from dog to deer despite similarity of 0.995.

CLIP’s classifiers shared the same image encoder, but their agreement fell from 97.4% on clean images to 81.5% on conflicts and 86.2% after shuffling. The CLIP head also achieved 97.2% clean accuracy with only 14.2% mean maximum confidence. Confidence magnitude depends on logit scale and should not be treated as a direct measure of classifier quality.

For visualization, features were L2-normalized and reduced to 50 dimensions using PCA. One joint t-SNE projection was fitted per backbone using clean and transformed features, with perplexity 30, 1,000 iterations, and seed 6304. The plots show greater class mixing among CLIP’s shuffled features, consistent with its larger accuracy loss. Projected distances are qualitative and should not replace measurements in the original feature space.

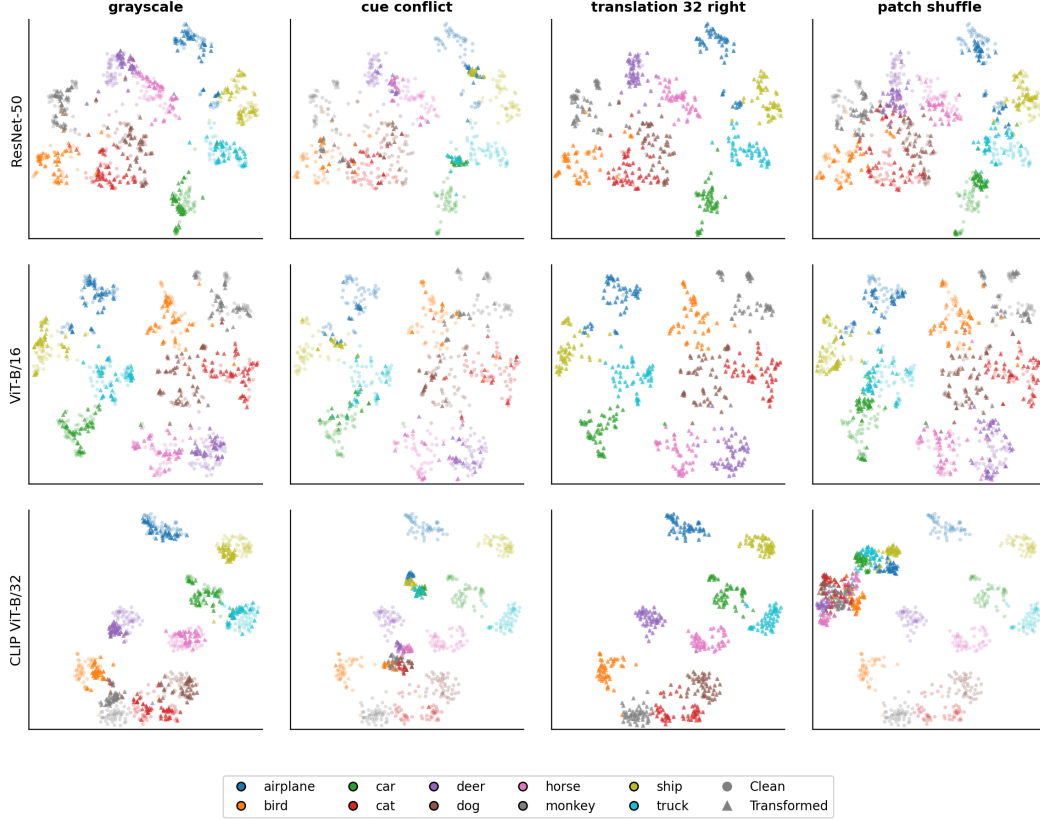


Figure 3: Joint t-SNE projections of clean and transformed representations. Rows show backbones; columns show the transformations. Colors identify classes while circles and triangles distinguish clean and transformed examples.

Architecture and Limitations (RQ4): ResNet’s translation consistency is compatible with convolutional structure, while ViT showed stronger shape preference on these conflicts. However, architecture is confounded with pretraining data, supervision, augmentation, and capacity. These findings characterize the evaluated systems rather than universal CNN or Transformer behavior.

Cue-conflict acceptance relied on subjective manual review. The extended review process may have introduced inconsistency in applying the criteria, so some retained images may provide ambiguous competing cues. Shape-bias results should therefore be interpreted cautiously.

3 Task 2: Unsupervised Domain Adaptation

Setup and training changes: Photo, Art Painting, and Cartoon served as labeled PACS sources, with Sketch as the unlabeled target. The comparison included Source-only, DAN, DANN Ganin et al. [2016], and CDAN Long et al. [2018]. Checkpoints were selected using mean source-validation macro-F1.

Original DANN showed exploding losses, while CDAN assigned 98.83% of target predictions to house. Recovery runs therefore started from the trained Source-only checkpoint. Learning rates were reduced from 10^{-4} to 10^{-5} ; DAN-10 used 10^{-6} for its backbone. DANN/CDAN maximum gradient reversal decreased from 1 to 0.1. Further changes included limiting gradient norms to 1, freezing BatchNorm scale and bias parameters as well as running statistics, and increasing alignment strength gradually over five epochs. The epoch budget changed from 30 to 20, early-stopping patience from 5 to 8, and checkpoint selection began at epoch 5.

These changes followed inspection of original target results, so recovery runs are exploratory follow-ups. The original four-method comparison is retained separately.

Source-to-target gap (RQ1): Source-only achieved 93.86% mean source accuracy and 69.15% Sketch accuracy, a descriptive gap of 24.71 percentage points. Dog and person accuracies were 34.07% and 45.63%, compared with 99.19% for elephant. The model classified 356 dogs and 214 horses as elephants, showing that the domain shift affected some classes more severely than others.

Table 3: Original and recovery results (%). Photo, Art, Cartoon, Mean, and Sketch cells report accuracy/macro-F1. Δ is Sketch accuracy change from Source-only in percentage points. Probe is held-out source-target domain accuracy; chance is 50%. Original DAN uses $\lambda = 1$; recovered DAN uses $\lambda = 10$.

Method	Photo	Art	Cartoon	Mean	Sketch	Δ	Probe
Original runs							
Source-only	96.7/96.0	90.0/90.0	94.9/95.2	93.86/93.74	69.15/69.73	0.00	99.86
DAN	97.0/96.5	90.0/90.1	94.9/95.2	93.96/93.94	59.20/53.84	-9.95	82.42
DANN	29.6/16.4	22.2/14.1	19.8/17.6	23.89/16.04	18.05/6.37	-51.11	99.86
CDAN	53.0/35.2	40.5/31.7	39.2/31.3	44.24/32.76	2.09/0.65	-67.07	99.18
Exploratory recovery runs							
DAN-10	93.7/92.8	81.2/81.2	90.8/91.1	88.59/88.34	65.11/61.76	-4.05	98.63
DANN	96.7/96.1	89.0/88.6	91.7/92.3	92.47/92.34	22.30/12.02	-46.86	100.00
CDAN	97.3/96.9	94.1/94.0	95.7/96.0	95.73/95.62	79.38/82.86	+10.23	99.73

Transfer and domain separability (RQ2): The changed settings improved source classification, but did not help every model recognize sketches. CDAN exceeded Source-only target accuracy by 10.23 points. DANN achieved 92.47% source accuracy but only 22.30% on Sketch. DAN-10 improved from 4.07% to 65.11% on Sketch, but still performed worse than Source-only.

Original DAN reduced the probe score from 99.86% to 82.42%, but Sketch recognition became worse. Recovered CDAN improved recognition even though its domain score remained 99.73%. Being able to distinguish between domains does not automatically make objects easier to classify.

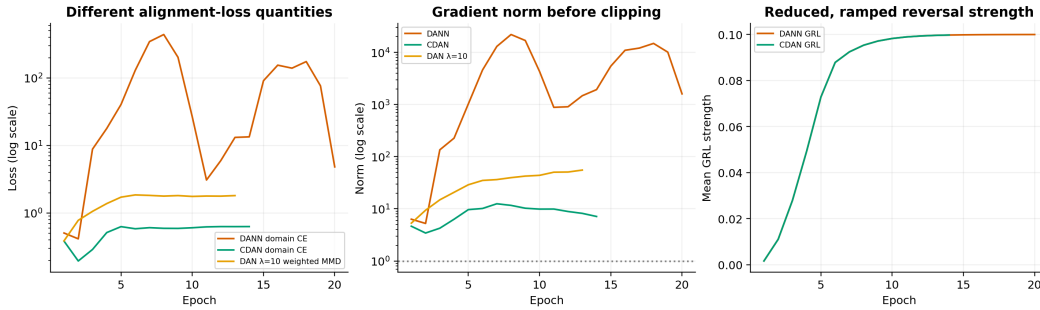


Figure 4: Recovery alignment and optimization diagnostics. The discriminator used during training is different from the independent held-out domain probe reported in the results table.

Conditional versus marginal alignment (RQ3): DANN aligns source and target features without directly including predicted class information. CDAN also uses class predictions, which may help align examples from corresponding classes. Under otherwise matched recovery settings, CDAN performed much better than DANN.

Compared with Source-only, CDAN improved person, dog, and horse accuracy by 52.50, 38.73, and 24.88 points. However, giraffe and elephant accuracy fell by 23.51 and 12.16 points. DANN still assigned 57.01% of sketches to horse.

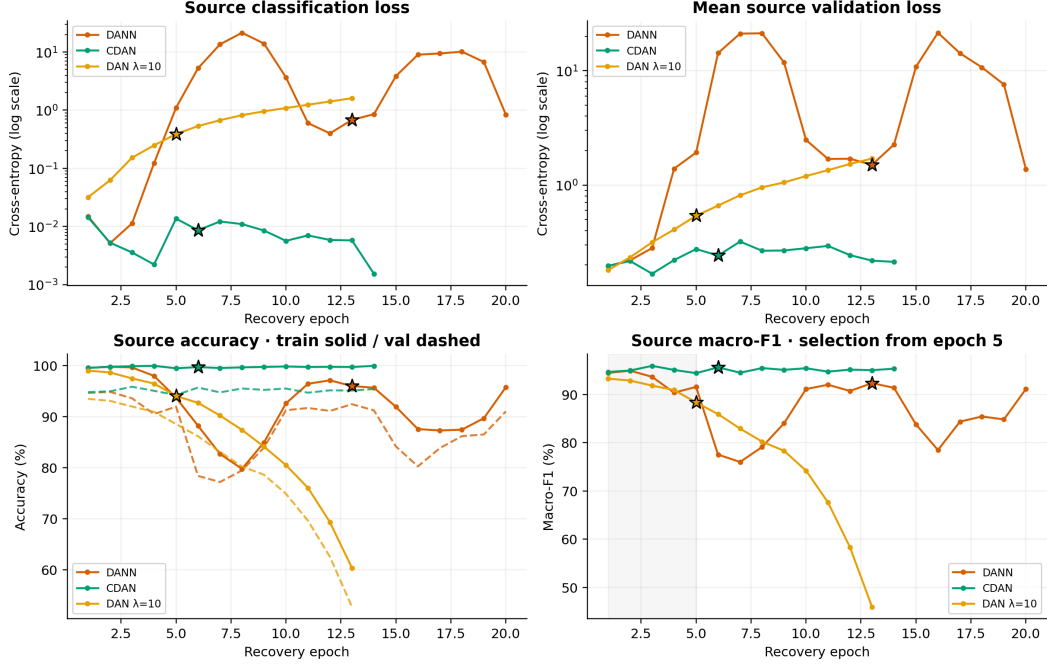


Figure 5: Recovery training and source-validation curves. Stars identify selected checkpoints.

Controlled alignment-strength study (RQ4): The original DAN study changed only λ , which controls how strongly training tries to align the domains. The idea is that stronger alignment might reduce domain differences, but too much could remove information needed to recognize classes.

As λ increased, the domains became harder to distinguish, but classification became worse. At $\lambda = 10$, the model predicted person for every sketch.

The $\lambda = 0.1$ setting had the highest source-validation macro-F1, so it could have been chosen without checking target labels. Its Sketch accuracy was only 0.28 points above Source-only, while its target macro-F1 was lower. The recovered DAN-10 result is kept separate because its starting checkpoint and training settings also changed.

Table 4: Original controlled DAN study (%). Only λ varies. Source metrics average the three validation domains. Recovered DAN-10 uses different training settings.

λ	Source acc.	Source F1	Probe	Target acc.	Target F1
0.1	95.54	95.61	92.72	69.43	65.31
1	93.96	93.94	82.42	59.20	53.84
10	21.66	5.07	73.49	4.07	1.12

4 Task 3: Domain Generalization

Setup and recovery changes: Task 2’s Source-only checkpoint was used as the ERM baseline. Original Task 3 training and model selection used only Photo, Art Painting, and Cartoon, excluding Sketch. DAN-DG used MMD to make features from the source domains more similar, with $\lambda = 1$ Long et al. [2015]. SAM used $\rho = 0.05$ to encourage low classification loss even after small changes to model parameters Foret et al. [2021].

Original DAN-DG failed to learn useful classification. Recovery runs therefore started from ERM and reduced the learning rate from 10^{-4} to 10^{-5} . BatchNorm scale and bias parameters were frozen, gradient norms were limited to 1, and alignment strength increased gradually over five epochs. The epoch budget changed from 30 to 20, early-stopping patience from 5 to 8, and checkpoint selection

began at epoch 5. Recovered DAN-DG with $\lambda = 10$ used a backbone learning rate of 10^{-6} . DAN-DG does not use gradient reversal.

Source validation versus Sketch (RQ1): SAM improved average source performance and performance on the weakest source domain. It also increased Sketch accuracy by 5.34 percentage points over ERM. However, recovered DAN-DG had slightly better average and weakest-domain source accuracy than SAM, but performed worse on Sketch.

Compared with ERM, SAM improved person and dog accuracy by 28.13 and 11.14 points, but elephant accuracy fell by 7.43 points. Dog accuracy remained only 45.21%, with 222 dogs classified as elephants.

Table 5: Task 3 results. Paired values report accuracy/macro-F1 (%). D and R denote original and recovered DAN-DG, both with $\lambda = 1$. Δ is Sketch accuracy change from ERM. The source-domain probe has 33.3% chance accuracy. Sharpness measures the validation-loss increase after a parameter perturbation of radius 0.05.

Metric	ERM	D	SAM	R
Photo	96.7/96.0	25.7/5.9	97.3/96.7	97.3/97.0
Art	90.0/90.0	22.0/5.1	92.0/91.9	93.2/93.4
Cartoon	94.9/95.2	17.3/4.2	96.4/96.8	95.5/96.0
Mean source	93.86/93.74	21.66/5.07	95.21/95.13	95.33/95.46
Worst source	90.00/90.01	17.27/4.21	91.95/91.86	93.17/93.38
Sketch	69.15/69.73	4.07/1.12	74.50/76.21	69.18/69.10
Sketch Δ	0.00	-65.08	+5.34	+0.03
Source probe	86.38	54.49	84.05	–
Sharpness	0.2333	0.0152	0.1202	–

Source alignment and classification (RQ2): A separate classifier tested whether features came from Photo, Art Painting, or Cartoon. Original DAN-DG reduced its accuracy from 86.38% to 54.49%, meaning the source domains became harder to distinguish. However, object classification also failed since source accuracy fell to 21.66%, and every sketch was classified as person. Classification loss remained near 1.93 even as MMD decreased.

Recovery restored source accuracy to 95.33%, but Sketch accuracy only matched ERM. Person and dog accuracy improved by 38.75 and 11.14 points, while giraffe and elephant declined by 13.01 and 5.68 points. Recovery fixed the severe training failure, but did not improve overall Sketch accuracy meaningfully beyond the baseline.

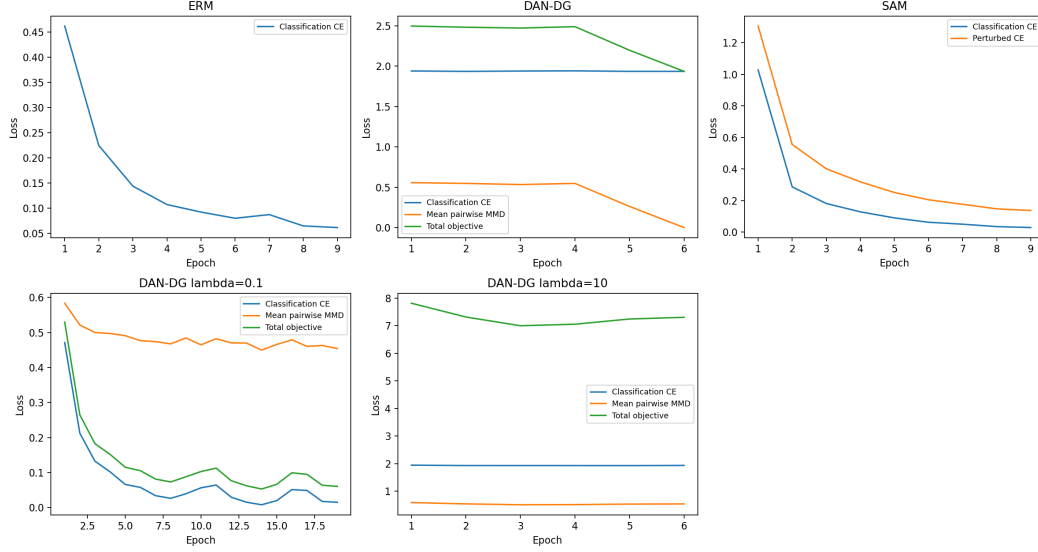


Figure 6: Original training losses. DAN-DG at $\lambda = 1$ and 10 keeps a high classification loss. ERM, SAM, and DAN-DG at $\lambda = 0.1$ learn the source classification task.

Sharpness and Sketch performance (RQ3): The sharpness test measures how much validation loss increases after the same small parameter perturbation. SAM reduced this increase from ERM’s 0.2333 to 0.1202 and also improved Sketch accuracy.

However, failed DAN-DG had an even smaller increase, 0.0152, despite only 4.07% Sketch accuracy. Its loss was already high before the perturbation, approximately 1.93. A model that performs poorly can remain poor after a small change, producing a low sharpness score.

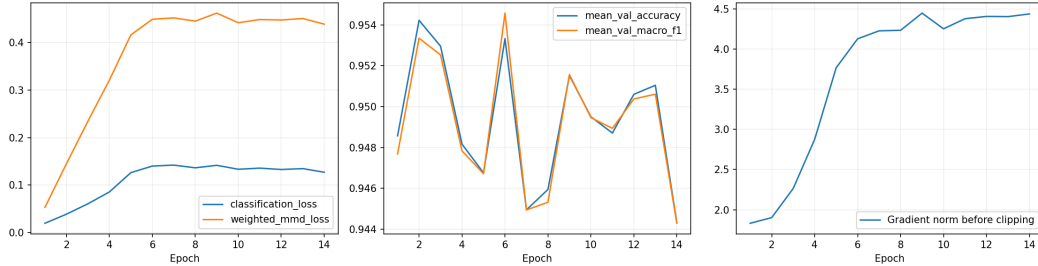


Figure 7: Recovered DAN-DG ($\lambda = 1$): classification and weighted MMD losses, source-validation scores, and gradient norms before clipping. The weighted MMD loss also reflects the gradual increase in alignment strength.

Controlled alignment study: The original study changed only λ , which controls the importance of source alignment. The weakest setting, $\lambda = 0.1$, had the highest source macro-F1 and could have been selected without checking Sketch labels. However, its Sketch accuracy remained below ERM.

At $\lambda = 1$ and 10, classification failed. Increasing λ also did not consistently make domains harder to distinguish. Probe accuracy was higher at 10 than at 1.

Recovered $\lambda = 10$ reached only 15.27% Sketch accuracy. It is kept outside this controlled comparison because its starting checkpoint and training settings also changed.

Table 6: Original DAN-DG alignment-strength study (%). Source F1 is averaged across source validation domains. Only λ changes between these runs.

λ	Source F1	Source probe	Sketch acc.	Sketch F1
0.1	95.14	71.10	62.03	64.22
1	5.07	54.49	4.07	1.12
10	5.07	67.11	4.07	1.12

Does access to Sketch help? (RQ4): Task 2 DAN sees unlabeled sketches during training; Task 3 DAN-DG does not. At the original $\lambda = 1$ setting, DAN achieved 59.20% Sketch accuracy, compared with DAN-DG’s 4.07%. However, DAN-DG also failed on source images, so this difference cannot be explained by target access alone.

At $\lambda = 0.1$, both models learned the source task well. DAN achieved 69.43% Sketch accuracy, compared with 62.03% for DAN-DG. This suggests that seeing unlabeled target images can help, but the methods also align different distributions. DAN aligns sources with Sketch, while DAN-DG aligns sources with each other.

5 Task 4: Open-Set Recognition

Setup: Models were trained on CIFAR-10 and tested for unknown rejection using the prescribed near and far CIFAR-100 classes, with 800 images per group. Near unknowns resemble known classes; far unknowns are less closely related. CIFAR-100 images were used only for final evaluation. Rejection thresholds were chosen to accept 95% of known validation images. Test acceptance was measured separately.

Four rejection scores were compared on the same Vanilla model. Vanilla, GCSC, and PROSER were then compared using MLS Vaze et al. [2022]. GCSC used RandAugment with two operations and magnitude 9. PROSER used additional classifier outputs and mixed features to represent possible unknowns Zhou et al. [2021]. Its known-class accuracy was calculated using only the ten known-class outputs.

Semantic similarity and failures (RQ1): Vanilla MLS rejected 30.50% of near unknowns and 54.25% of far unknowns. Pickup trucks, wolves, and buses were incorrectly accepted at rates of 90%, 88%, and 83%, respectively. However, unrelated objects were also accepted. 60% of wardrobes and 56% of bowls and mushrooms passed the rejection threshold.

The failure examples include three near and three far unknowns. Bus-to-truck and pickup-to-automobile mistakes are understandable because the objects share vehicle features. Fox-to-dog is also visually and semantically plausible. Mushroom-to-frog, bowl-to-cat, and wardrobe-to-cat are more surprising. Color and texture may contribute at this low resolution, but the images alone cannot confirm why these predictions occurred.

What the scores measure (RQ2): MSP measures the highest softmax confidence. MLS uses the largest raw class score before softmax. Energy combines evidence from all class scores. Mahalanobis measures how far an image’s features are from known-class feature distributions. These methods use different signals to decide whether an image is unfamiliar.

AUROC measures how well a score separates known and unknown images across thresholds. Rejection rate measures performance at the chosen threshold. MSP had the highest near AUROC, but rejected fewer near unknowns than MLS at that threshold. Mahalanobis had the highest far AUROC, while Energy rejected the most far unknowns. A better AUROC did not always mean better rejection at the selected threshold.

MLS and Energy behaved similarly: their near-unknown rank correlation was 0.998, and their accept/reject decisions differed on only 1.63% of near images. MLS and Mahalanobis disagreed on 13.13%. Feature distance gave different information, but was not consistently better than class scores.

Table 7: Rejection scores on the same Vanilla model. Values are percentages. N/F/A denote near, far, and all unknowns. K is known-test acceptance. Every threshold targets 95% known-validation acceptance. Vanilla known-class accuracy is 94.83%.

Score	AUROC			Rejection		
	N	F	A	N	F	K
MSP	80.55	89.52	85.03	25.38	40.13	95.34
MLS	78.72	89.72	84.22	30.50	54.25	95.02
Energy	78.74	89.81	84.28	30.63	54.63	95.03
Mahalanobis	79.37	91.13	85.25	26.13	48.13	95.02

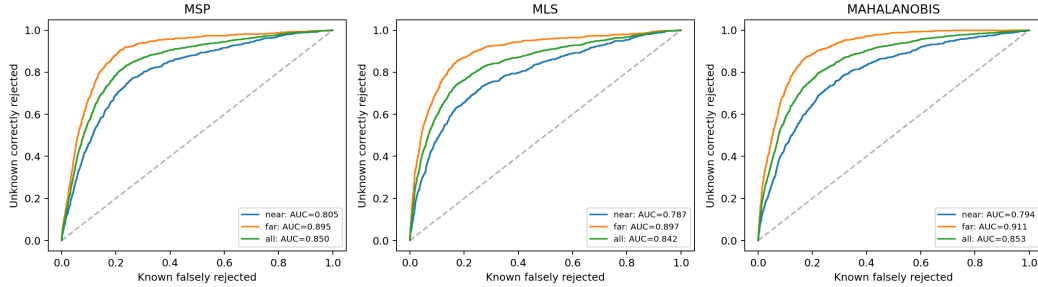


Figure 8: Vanilla ROC curves for MSP, MLS, and Mahalanobis. The curves show the trade-off between rejecting unknown images and incorrectly rejecting known images as the threshold changes. Far unknowns are generally easier to distinguish.

Augmentation and rejection (RQ3): GCSC increased known-class accuracy slightly, from 94.83% to 94.98%. Near and far rejection improved by 5.25 and 3.50 percentage points, and AUROC increased for both groups.

Training on varied versions of known images may help the model learn features that are less dependent on narrow visual patterns. However, this experiment does not identify which augmentation caused the gains. Even with GCSC, near unknowns remained harder to reject than far unknowns.

Table 8: Training-method comparison (%). MLS is used for all three models; PH is PROSER’s additional placeholder score. CSA is known-class accuracy before rejection, and K is known-test acceptance. N/F/A report near/far/all-unknown AUROC; R_N/R_F report near/far rejection.

Model / score	CSA	K	N	F	A	R_N	R_F
Vanilla / MLS	94.83	95.02	78.72	89.72	84.22	30.50	54.25
GCSC / MLS	94.98	94.90	81.63	91.11	86.37	35.75	57.75
PROSER / MLS	94.25	94.72	78.84	86.89	82.87	28.13	43.75
PROSER / PH	94.25	94.97	74.05	84.27	79.16	25.25	39.88

Placeholder learning (RQ4): PROSER adds classifier outputs that provide an alternative to the known classes. It also mixes features from different known classes to create artificial unknown examples. The aim is to teach rejection without using real unknown images during training.

In this run, PROSER reduced known-class accuracy by 0.58 percentage points. With MLS, near and far rejection fell by 2.37 and 10.50 points compared with Vanilla. Near AUROC increased only slightly, while far AUROC fell. Its own placeholder score gave still lower rejection rates and AUROC.

One possible limitation is that mixtures of known-class features may not resemble real unknowns. Learning to reject these mixtures does not necessarily teach the model to reject unfamiliar objects elsewhere in feature space.



Figure 9: Three near and three far unknowns incorrectly accepted by Vanilla MLS. Each panel shows the unknown class, predicted known class, score, and threshold. The unknownness score is $u = -\max_k z_k$; an image is accepted when $u \leq \tau$, with $\tau = -5.7620$. All six unknown images incorrectly pass this rule.

6 Conclusion

The experiments show that robustness depends on more than clean classification accuracy. ResNet, ViT, and CLIP exhibited different visual sensitivities, and representation stability did not always imply prediction stability. Domain alignment sometimes damaged class discrimination, while low sharpness alone did not guarantee useful generalization.

SAM improved Sketch recognition, and recovered CDAN produced promising exploratory adaptation results. In open-set recognition, augmentation improved rejection, whereas PROSER offered no broad advantage over the vanilla model. These findings highlight the need to evaluate classification, transfer, and unknown rejection separately.

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